



Machine Learning for High Dimensional Data Analysis

Acknowledgements

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- Sam Holt
- Andrea Petrocchi

Supervisors :

- Hans Fangohr
- Guido Meier

Outline

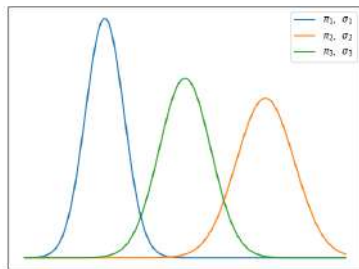
- 1 Problem statement
- 2 Clustering
- 3 Dimensionality Reduction
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- 7 Results for Dataset 2
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Problem statement

- Micromagnetic simulation-based parameter space exploration generates large datasets.
- Manual classification of equilibrium states is time-consuming.
- Machine learning-based automation would save a lot of time and effort.

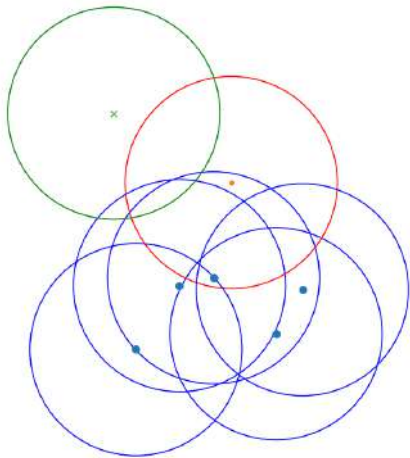
Clustering

Gaussian Mixture Models(GMM)



$$p(\mathbf{x}) = \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$$

DBSCAN



$\epsilon = 0.5$, $\text{min_pts} = 3$

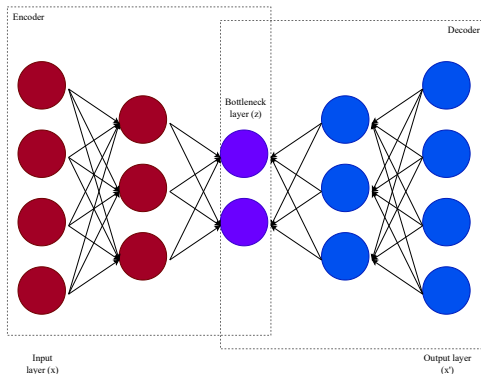
- ϵ : The radius around a point that defines the neighborhood.
- min_pts : Minimum number of samples required to call a region dense.

Curse of Dimensionality

- Uninformative Distance and Similarity Measures
- Data sparsity
- Difficulty in Outlier Assessment

Dimensionality Reduction

Autoencoder



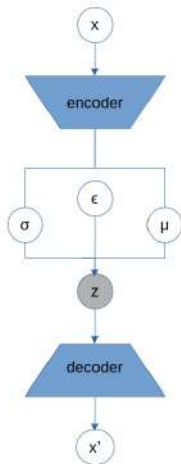
- 1 Input space : $\mathcal{X} = \mathbb{R}^n$
- 2 Latent space : $\mathcal{Z} = \mathbb{R}^m$,
 $m < n$
- 3 Encoder :
 $f_{\theta} : \mathcal{X} \rightarrow \mathcal{Z}, z = f_{\theta}(x)$
- 4 Decoder :
 $g_{\phi} : \mathcal{Z} \rightarrow \mathcal{X}, x' = g_{\phi}(z)$
- 5 Mean Squared Error Loss :
$$\text{MSE}(x, x') = \frac{1}{n} \sum_{i=1}^n (x - x')^2$$

Probabilistic vs deterministic modelling

	Deterministic Models	Probabilistic Models
Definition	Produces a fixed output for a given input, $f_{\theta}(x)$	Produces a probability distribution over possible outputs, $p_{\theta}(y x)$.

- Handling Uncertainty
- Flexibility of Cluster Shapes
- Robustness to Noise
- Generation of new data

Variational Autoencoder(VAE)



- 1 Pass x through the encoder,
 $q(z|x) = \mathcal{N}(z|\sigma_x, \mu_x^2)$
- 2 Sample noise, $\epsilon \sim \mathcal{N}(0, 1)$
- 3 Reparametrize, $z = \epsilon\sigma_x + \mu_x$
- 4 Pass z through the decoder, x_r
- 5 Compute reconstruction loss,
 $\mathcal{L}_{\text{rec}} = \|x - x_r\|^2$
- 6 Compute KL divergence loss,
 $\mathcal{L}_{\text{KL}} = D_{\text{KL}}(\mathcal{N}(\mu_x, \sigma_x^2) \parallel \mathcal{N}(0, I))$
- 7 Compute total VAE loss,
 $\mathcal{L} = \mathcal{L}_{\text{rec}} + \mathcal{L}_{\text{KL}}$

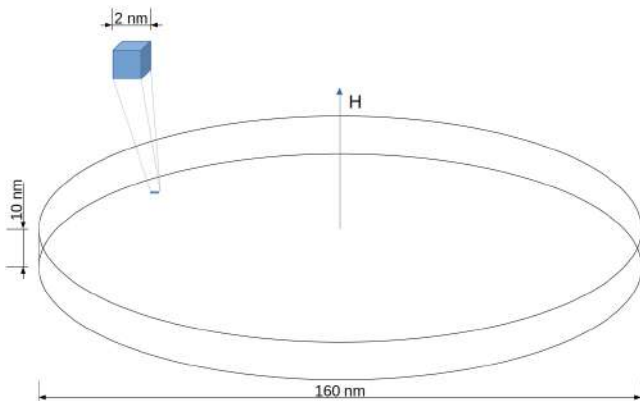
Evidence Lower Bound(ELBO) Optimization

$$\mathcal{L}_{\theta,\phi}(x) = \underbrace{E_{q_{\phi}(z|x)}[\log p_{\theta}(x|z)]}_{\text{reconstruction error}} - \underbrace{D_{KL}(q_{\phi}(z|x)||p_{\theta}(z))}_{\text{regularization term}}$$

- Expected Log-Likelihood(reconstruction error) : Biases the model towards configurations of latent variables that explain the observed data.
- Negative KL Divergence between approximate posterior and prior (regularization term) : Encourages the approximate posterior $q_{\phi}(z|x)$ to stay close to the prior distribution $p_{\theta}(z)$ over the latent variables. Avoids $q_{\phi}(z|x)$ collapsing to a point estimate, leading to poor generalization.

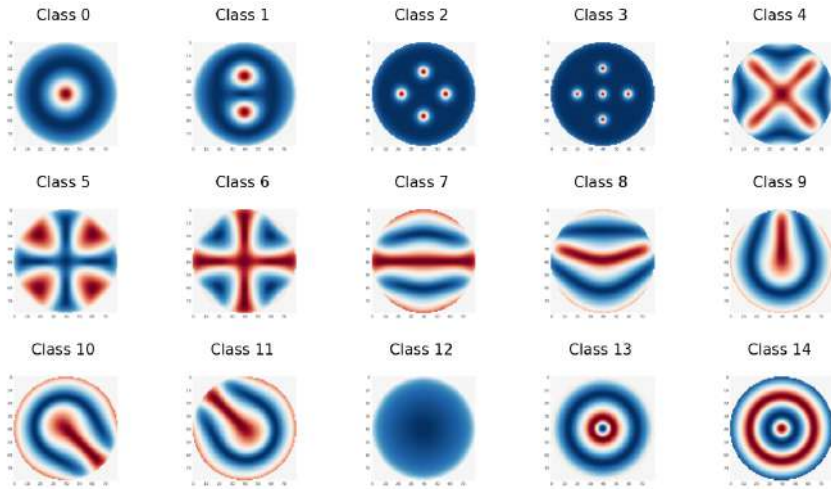
Dataset 1

Dataset 1



- FeGe disc
- 345 randomly chosen field strengths between 0 and $3.18 \times 10^9 \text{ Am}^{-1}$.
- A total of 3010 equilibrium states.
- Only the center slice is considered.

Magnetization configurations

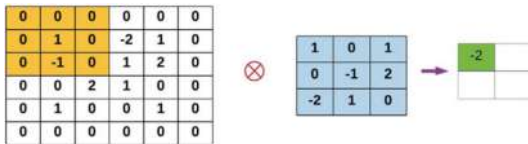


Convolutional blocks

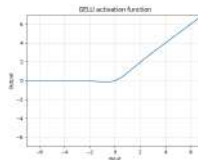
- Convolutional layer
- Activation function
- Batch Normalization

Why CNN?

- Capture spatial information
- Computationally efficient
- Locality assumption



a

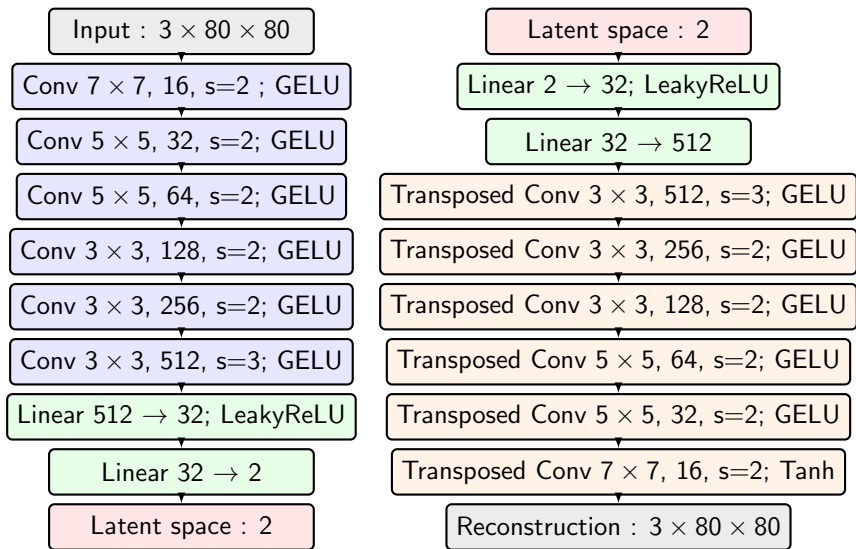


b

a

^b<https://docs.pytorch.org/docs/stable/generated/torch.nn.ReLU.html>

Network Architecture for Dataset 1



Results for Dataset 1

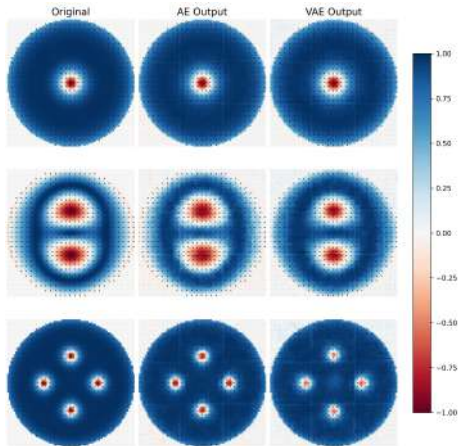


Figure: Comparison of original and reconstructed field images, Dataset 1

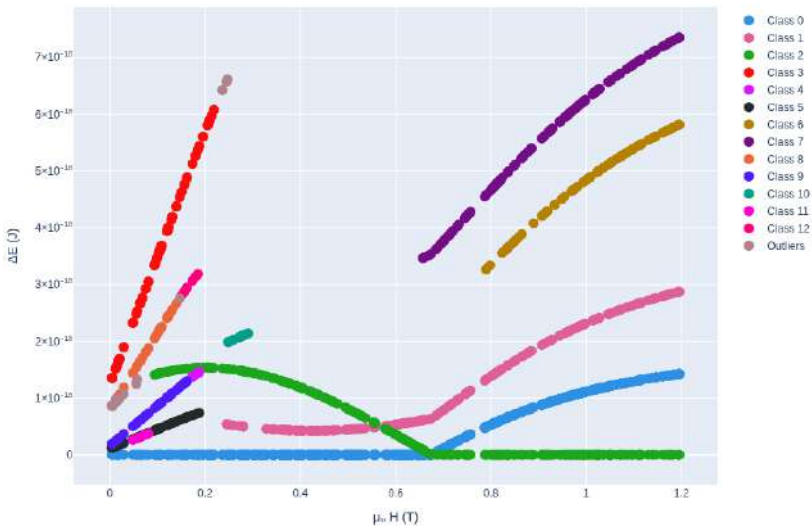
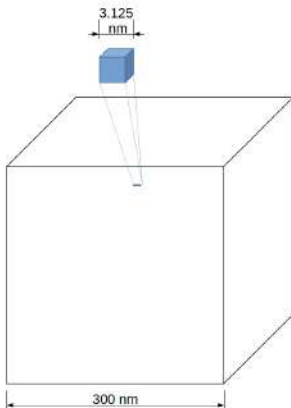


Figure: DBSCAN Clustering(VAE), Dataset 1, $\epsilon = 0.5$, $\text{min_pts} = 5$

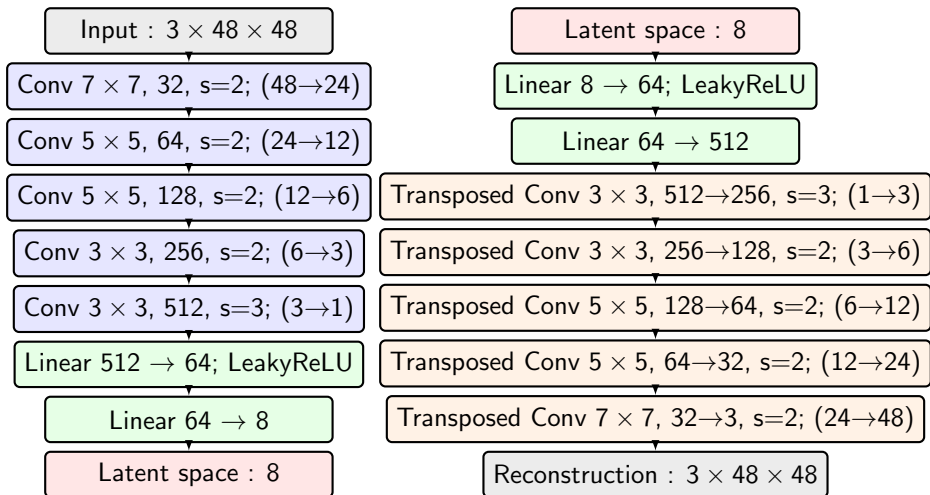
Dataset 2

Dataset 2

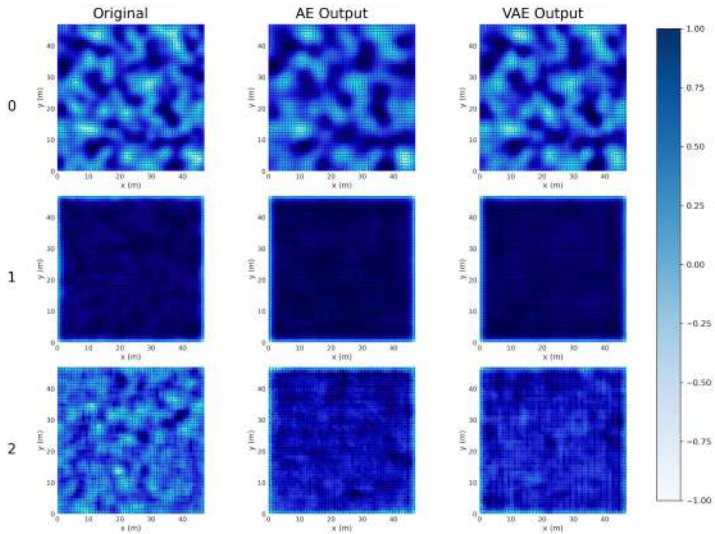


- FeGe cube
- Magnetic field strength : 0 to $0.39 \times 10^6 \text{ Am}^{-1}$.
- Temperature : 0 to 3000 K.
- A total of 2601 equilibrium states.

Network Architecture for Dataset 2



Results for Dataset 2



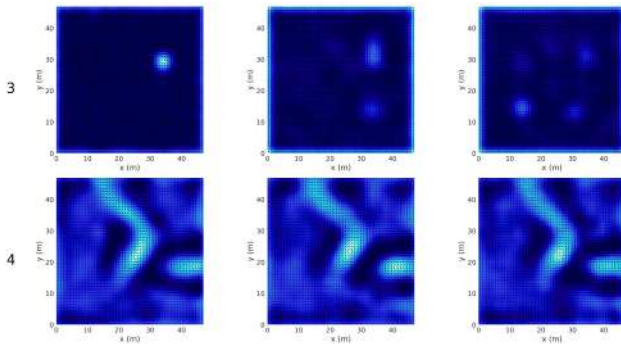


Figure: Comparison of original and reconstructed field images, Dataset 2

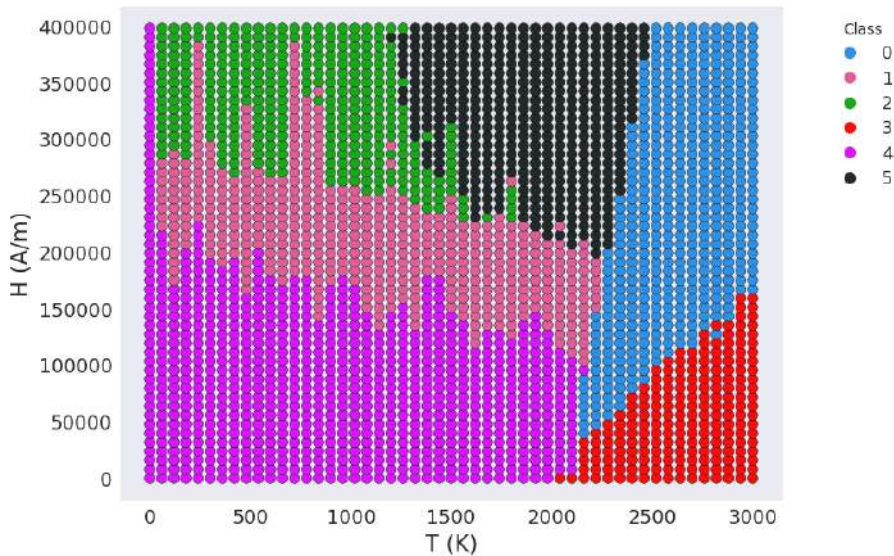


Figure: GMM Clustering(AE), Dataset 2.

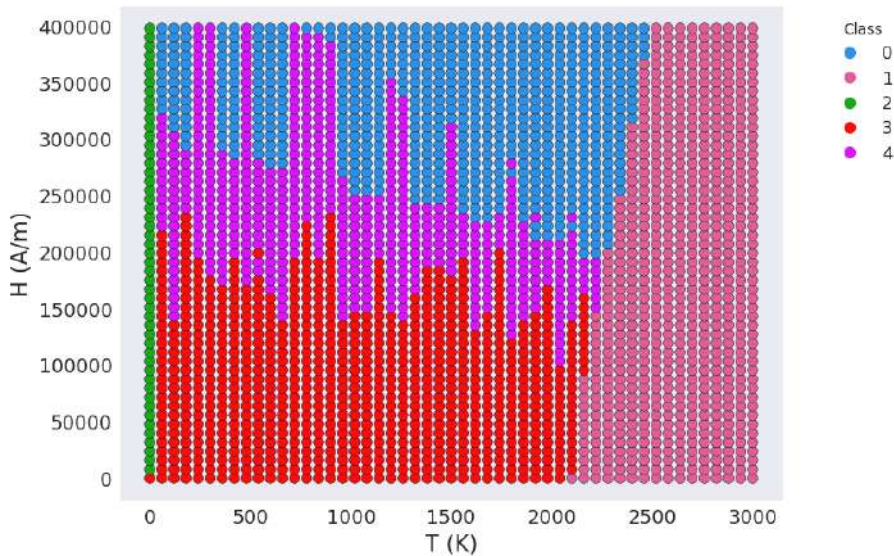


Figure: GMM Clustering(VAE), Dataset 2.

External metric values

Dataset	Model	Adjusted Rand Index(ARI)	Purity
1	VAE	0.9990	0.9988
2	AE	0.6230	0.8812
2	VAE	0.8093	0.9234

Latent Space Exploration

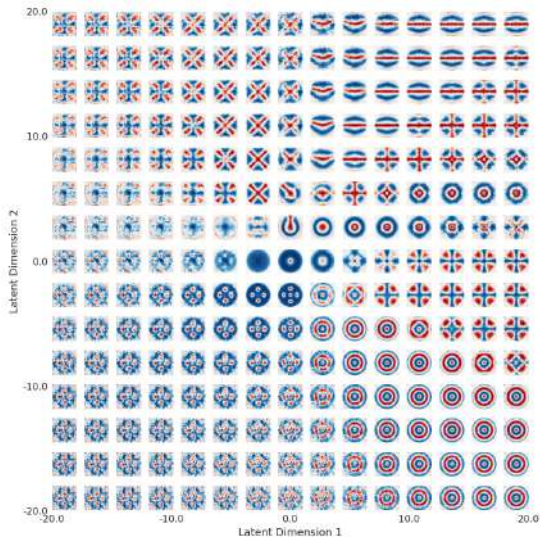
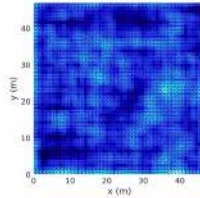
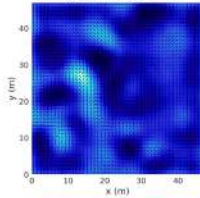


Figure: Magnetization configurations obtained by uniformly sampling from the VAE latent space

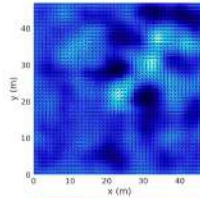
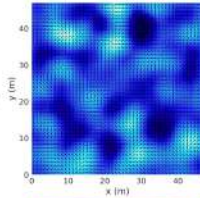
AE Output

VAE Output

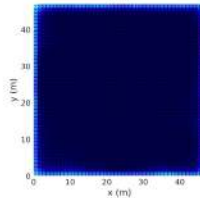
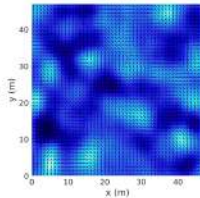
$\sigma=1$



$\sigma=2$



$\sigma=3$



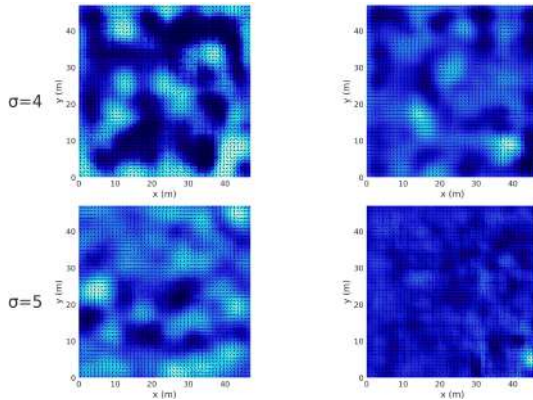


Figure: Perturbed input for Latent space exploration

Thank you!

In summary,

- Autoencoders were successful at reducing the dimensionality of micromagnetic data while preserving essential physical features.
- Clustering in latent space was effective in the identification of meaningful magnetic states.
- Variational Autoencoders enabled a structured and continuous latent space, improving clustering quality and allowing generative exploration of micromagnetic configurations.
- Combining representation learning with unsupervised clustering provides an effective and scalable framework for analyzing complex physical systems.

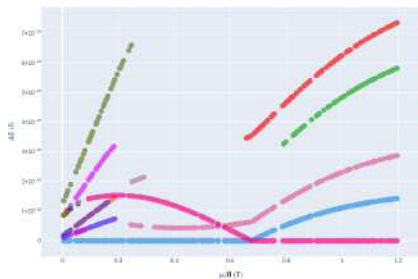
The software developed for this study is available as open-source code at

https://github.com/Sassy-Buoy/3D_periodic_clustering and

<https://github.com/Sassy-Buoy/>

[hyper-param-optim-4-vector-field-clustering](#).

Ground Truth



Class 0



Class 1



Class 2



Class 3



Class 4



Class 5



Class 6



Class 7



Class 8



Class 9



Class 10



Class 11



Class 12



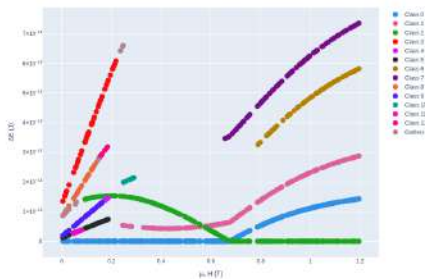
Class 13



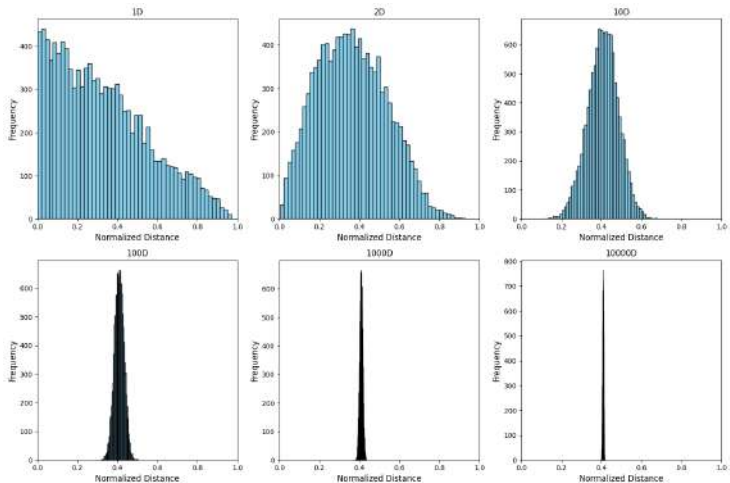
Class 14



DBSCAN Clustering

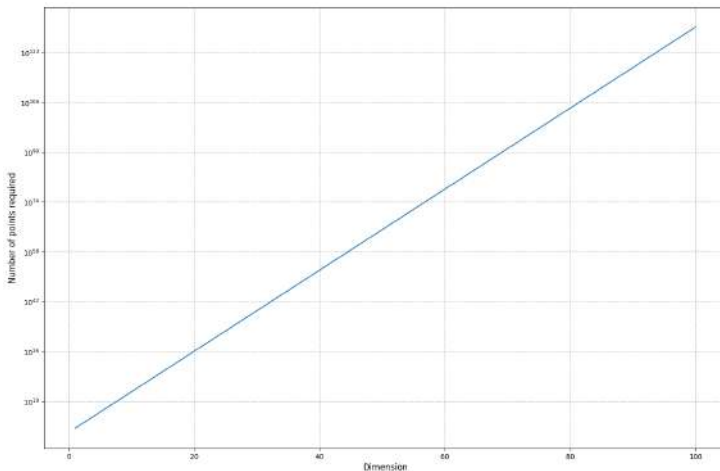


Uninformative Distance and Similarity Measures



Normalized Pairwise Distance Distributions in Different Dimensions

Data sparsity, Difficulty in Outlier Assessment



Points required to uniformly sample space (5% per dimension)



Expectation Maximization(EM) algorithm

- ① **Expectation Step (E-Step):** Compute the responsibilities:

$$\gamma_{ik} = \frac{\pi_k \mathcal{N}(\mathbf{x}_i \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_{j=1}^K \pi_j \mathcal{N}(\mathbf{x}_i \mid \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)}$$

- ② **Maximization Step (M-Step):** Update the parameters based on the responsibilities:

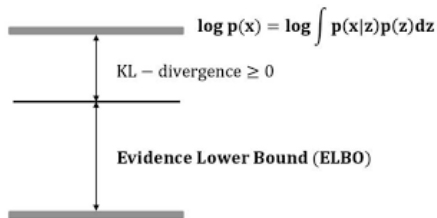
$$\pi_k = \frac{N_k}{N}, \quad \boldsymbol{\mu}_k = \frac{1}{N_k} \sum_{i=1}^N \gamma_{ik} \mathbf{x}_i$$

$$\boldsymbol{\Sigma}_k = \frac{1}{N_k} \sum_{i=1}^N \gamma_{ik} (\mathbf{x}_i - \boldsymbol{\mu}_k)(\mathbf{x}_i - \boldsymbol{\mu}_k)^\top$$

where $N_k = \sum_{i=1}^N \gamma_{ik}$ and N is the number of data points.

Variational Autoencoder and Variational inference

$$p_{\theta}(z|x) \approx q_{\phi}(z|x)$$



1

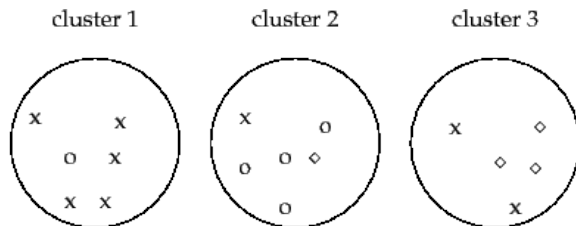
¹<https://cedar.buffalo.edu/~srihari/CSE676/19.4%20VariationalInference.pdf>

Kullback–Leibler(KL) divergence

KL divergence is a statistical measure used to compare probability distributions. Given a true probability distribution P defined over a random variable x and a model probability distribution Q defined over the same variable,

$$D_{\text{KL}}(P||Q) = \begin{cases} \sum_{x \in X} p(x) \log \frac{p(x)}{q(x)} & \text{for discrete distributions,} \\ \int_{-\infty}^{\infty} p(x) \log \frac{p(x)}{q(x)} dx & \text{for continuous distributions} \end{cases}$$

External Metrics : Purity



► **Figure 16.1** Purity as an external evaluation criterion for cluster quality. Majority class and number of members of the majority class for the three clusters are: x, 5 (cluster 1); o, 4 (cluster 2); and o, 3 (cluster 3). Purity is $(1/17) \times (5 + 4 + 3) \approx 0.71$. 2

$$\text{Purity} = \frac{1}{N} \sum_k \max_j |C_k \cap L_j|$$

²<https://nlp.stanford.edu/IR-book/html/htmledition/evaluation-of-clustering-1.html> 2/44

External Metrics : Rand Index and Adjusted Rand Index

	Same cluster	Different cluster
Same class	TP	FN
Different class	FP	TN

$$\text{Rand Index, } R = \frac{TP + FN}{TP + FP + FN + TN}$$

Rand Index takes values in the range of 0 to 1.

$$\text{Adjusted Rand Index, } R' = \frac{R - \bar{R}}{1 - \bar{R}}$$

where $\bar{R}$ is the expected R for random clusterings.

Adjusted Rand Index takes values in the range of -1 to 1 .

Variational Autoencoder and Variational inference

$$\underbrace{p_{\theta}(z|x)}_{\text{posterior}} = \frac{\overbrace{p_{\theta}(x|z)}^{\text{likelihood}} \overbrace{p_{\theta}(z)}^{\text{prior}}}{\underbrace{p_{\theta}(x)}_{\text{marginal}}}$$

Marginal distribution is intractable. Hence, the posterior distribution is approximated using variational inference.

$$p_{\theta}(z|x) \approx q_{\phi}(z|x)$$
$$\log p_{\theta}(x) = \underbrace{E_{q_{\phi}(z|x)} \left[\log \frac{p_{\theta}(x, z)}{q_{\phi}(z|x)} \right]}_{\text{ELBO}} + \underbrace{E_{q_{\phi}(z|x)} \left[\log \frac{q_{\phi}(z|x)}{p_{\theta}(z|x)} \right]}_{\text{KL Div}}$$

$$D_{KL}(q_{\phi}(z|x) || p_{\theta}(z|x)) \geq 0 \implies \mathbf{L}_{\theta, \phi}(\mathbf{x}) \leq \log \mathbf{p}_{\theta}(\mathbf{x})$$

The KL divergence term represents the "gap" between the Evidence "Lower Bound" function and the true log marginal likelihood.